

# Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

## Education

### University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

**Coursework:** Systems for Data Science, Performance Engineering, Optimization in CS, Advanced Machine Learning, Advanced NLP

### Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

## Skills

### Languages

TypeScript, JavaScript, Python, Java, C++, SQL, Bash, HTML/CSS

### Frontend & Web

React, Redux, Electron, responsive UI, REST integration

### Backend & Data

Node.js, Express, FastAPI, Fastify, REST APIs, JWT authentication, rate limiting, async services, integration testing, PostgreSQL, MongoDB, Redis, SQLite

### Cloud & DevOps

AWS (SQS, KMS), GCP, Azure, Docker, Git, GitHub Actions, CI/CD, Linux

### AI Agents & Tools

LLM agents, tool use, provider routing / fallback, OpenAI / Groq APIs, Claude / Copilot / Cursor

## Experience

### Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a **full-stack data annotation platform (React, Node.js, MongoDB)** used daily by the AI team, replacing spreadsheet workflows with secure role-based REST APIs; cut manual work by **100+ hours/month** and improved throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**, integrated with Mayden's iaptus platform via **REST APIs** and **AWS SQS** messaging.
- Owned backend security: hardened **authentication and rate limiting**, implemented **AWS KMS** encryption for sensitive data, and remediated **20+ VAPT findings**, applying secure-coding and authorization practices across the stack.

### Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ REST APIs** in Node.js + Fastify backed by **PostgreSQL** with input validation and clear contracts, and migrated the codebase from **JavaScript to TypeScript** across multiple repositories.
- Integrated **LLM-powered** context-aware replies into a chat product, iterating on prompt design and response-quality heuristics for a real customer-facing experience used in production by the frontend team.

## Projects

### Underdogs Fitness - Production Full-Stack Platform

underdogsfitness.in

React, Redux, Node.js, Express, MongoDB Atlas, Stripe | solo, live in production 3+ years

- Built and deployed a **production full-stack platform** solo end-to-end (**React + Redux**, Node.js, Express, MongoDB Atlas) with **JWT-based authentication**, Stripe + manual-cash hybrid payments, and external integrations (Stripe, Firebase Storage, Vercel); moved **90%+** of payments through the portal.
- Designed for multi-branch from day 1 (every entity scoped by `branchId`, JWT-claim auth); when the client opened a 2nd branch a year later, onboarded it as a configuration change with **zero schema migration**; still live **3+ years** later.

### AgenticSearch - AI Agent with External Tool Integration

github.com/Deva-1903/ciir\_agentic\_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built an **AI agent** that integrates **external tools and APIs** (Brave Search, OpenAI / Groq) into a multi-stage pipeline, planning typed retrieval, extracting entities, verifying outputs, and returning cell-level provenance.
- Hardened the service with **150+ automated tests**, provider routing and fallback, and pipeline instrumentation; deployed as a live demo on **Google Cloud Run**.

### cf-ai-research-scout - Real-Time AI Agent (TypeScript)

github.com/Deva-1903/cf\_ai\_research\_scout

TypeScript, Cloudflare Workers, WebSockets, SQL

- Built an **AI agent in TypeScript** that fetches external web sources, runs multi-step tool use, and streams responses to the client over **WebSocket** in real time.
- Designed durable workflows so multi-step agent tasks **recover gracefully** from process restarts and tool failures, with state persisted server-side without an external database.

## Publication & Extracurriculars

- Publication:** "Alzheimer's Disease Classification using Transfer Learning," **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.